

Ashanti Hameed

Lead Data Engineer

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Hillside, NJ

Summary

Highly skilled **Lead Data Engineer** with over **12 years** of experience in designing, building, and optimizing large scale data pipelines and architectures across multiple industries, including **banking, payments, and retail**. Expertise in **cloud platforms** (AWS, GCP, Azure), **distributed data processing** (Apache Spark, Kafka, Dataflow), and **machine learning (ML)** technologies (TensorFlow, scikit-learn). Proven ability to lead data engineering initiatives, from **real-time data streaming** and **ETL pipeline** development to implementing **AI-driven solutions** that empower business intelligence and predictive analytics. Demonstrated leadership in building scalable, cost-effective cloud-based data platforms and driving **AI/ML strategies** that optimize data-driven decision-making. Adept at **collaborating** with cross-functional teams, mentoring engineers, and ensuring best practices in **data governance** and compliance. Passionate about continuous learning, leveraging emerging technologies to improve data quality and performance.

Skills

Data Engineering: ETL/ELT pipelines, data modeling, data warehousing, data lakes, real-time streaming, distributed systems, performance tuning, data quality enforcement.

Programming: Python, SQL, Scala, Bash

Big Data: Apache Spark, PySpark, Kafka, Flink, Databricks, Hadoop ecosystem

Cloud Platforms: AWS (S3, Lambda, Glue, Redshift, Kinesis), GCP (BigQuery, Dataflow, Pub/Sub), Azure (Data Factory, ADLS)

Data Warehousing: Snowflake, BigQuery, Redshift, Databricks Lakehouse, Postgres, MySQL, NoSQL

Orchestration: Airflow, dbt, Step Functions, Dagster

DevOps: Terraform, Docker, Kubernetes, Git, CI/CD pipelines, CloudWatch, Datadog

Governance & Security: RBAC, encryption/KMS, metadata management, GDPR/PCI compliance, data lineage

ML/AI: Feature engineering, ML data pipelines, scikit-learn, TensorFlow, MLflow

Tools: Looker, PowerBI, Tableau, Great Expectation, dbt tests

Experience

Nexus

Lead Data Engineer

07/2021 – Present

- Led the development and deployment of **end-to-end machine learning pipelines** using **Apache Spark** and **GCP Dataflow**, processing petabyte-scale transactional data.
- **Architected** a **cloud-based data platform** on **GCP**, using **BigQuery** and **Dataflow**, supporting large-scale, real-time analytics for fraud detection and customer insights.
- Developed and implemented **AI-powered analytics** for customer churn prediction and **credit risk assessment** using **TensorFlow** and **scikit-learn**.
- Integrated **real-time data streaming** solutions using **Apache Kafka** and **GCP Pub/Sub** for continuous, low-latency data ingestion.

- Designed and optimized **data pipelines** for both **batch and real-time processing**, improving model training and inference performance by 40%.
- Drove the implementation of **data governance protocols**, including **RBAC** and **audit logging**, ensuring compliance with **SOX** and **GDPR** regulations.
- **Managed** a team of **data engineers** and **data scientists**, providing technical leadership and mentoring on best practices for **cloud-native data architectures** and **ML model deployment**.

Key Technologies: GCP (BigQuery, Dataflow, AI Platform), Apache Kafka, TensorFlow, scikit-learn, Python, SQL, Google Pub/Sub, Apache Spark, Terraform, RBAC, CI/CD

Aditi Consulting

Senior Data Engineer | AI/ML Engineer

08/2016 - 06/2021

- **Led the architecture and implementation of data pipelines** integrating multiple data sources into **Snowflake**, enabling business intelligence and product analytics.
- Designed and **built real-time data streaming** solutions using **Kafka** and **AWS Kinesis**, providing up-to-date insights on **customer behavior** and **inventory levels**.
- Developed **machine learning models** for **demand forecasting** and **customer segmentation** using scikit-learn, TensorFlow, and **AWS SageMaker**.
- Spearheaded the development of **automated ETL workflows** using **dbt** and **Airflow**, improving operational efficiency and reducing manual intervention in data processes.
- Improved **data models** for predictive analytics by designing **data marts** and **reporting schemas** to enable deeper analysis and visualization.
- Drove the adoption of **CI/CD pipelines** using **GitLab CI** and **Terraform**, automating infrastructure deployment and scaling.
- Collaborated with **business intelligence** teams to develop **custom dashboards** for inventory management and marketing campaigns.

Key Technologies: Snowflake, dbt, Apache Kafka, AWS Kinesis, TensorFlow, scikit-learn, Python, SQL, Scala, AWS SageMaker, GitLab, CI, Terraform

Casper

Data Engineer | ML Engineer

05/2013 – 07/2016

- Joined a fast-growing **retail tech startup** as one of the early data engineers, helping build the company's **data ecosystem from scratch**, including ingestion frameworks, ETL pipelines, and data models.
- Design and developed **scalable ETL pipelines** for processing high-volume transactional data using **Apache Kafka** and **AWS Lambda**.
- Led the migration of legacy on-premise systems to a cloud-native **infrastructure** utilizing **AWS Redshift** and **S3** for centralized data storage and processing.
- Built **automated data validation** and **anomaly detection** pipelines using **Apache Airflow**, ensuring data quality for business-critical financial reports.
- Collaborated with **data scientists** to implement **fraud detection models** using scikit-learn, improving fraud detection accuracy by 15%.
- Optimized **data storage** and **query performance** on **AWS Redshift**, creating **data models** and partitioning strategies for faster aggregation.
- Worked with **cross-functional teams** to design **data-driven dashboards** for real-time fraud monitoring using **Power BI** and **Looker**.

- Mentored junior engineers on best practices for cloud migration, data pipeline design, and real-time streaming architectures.

Key Technologies: AWS (S3, Lambda, Redshift, Glue), Apache Airflow, Apache Kafka, Python, scikit-learn, PostgreSQL, SQL, Terraform

Education

Montclair State University

09/2010 – 03/2013

Bachelor of Arts in Psychology